

CHE 205, Fall 2019, Exam 2

Instructor: Shafigh Mehraeen

Due Date/Time: Fri, 11/08/2019, 9:55AM

Note:

- The exam is open-book and open-note. You are allowed to use internet as well; however, you are NOT allowed to talk to, text, or email anybody during the exam.
- Make sure you save your file frequently as excel may crash every minute!
- Also keep a copy of your work just in case the submission fails to transmit the file.

Problem 1 (10 pts). Consider the following 2nd order differential equation:

$$(1 - x^2) \frac{d^2V}{dx^2} + \alpha x \frac{dV}{dx} + \beta V = 0, \quad (1)$$

where $\alpha = -2$, and $\beta = 30$. The equation is subject to the following boundary conditions:

$$V = 0 \text{ at } x = 0, \quad \frac{dV}{dx} = -2 \text{ at } x = 0.5. \quad (2)$$

- (a) In a spreadsheet, solve above differential equation using forward Euler together with the shooting method. Use $\Delta x = 0.1$, $0 \leq x \leq 0.9$, and plot $V(x)$ against x in red (6 pts).
- (b) Solve above differential equation using the finite (central) difference approach, and plot $V(x)$ against x in blue on the same chart (4 pts).